Concretizing wickedness in problem- and project-based courses

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Abstract

This paper explores how wicked problems are perceived and operationalised in engineering education. Focusing on a particular technical university in Northern Europe, this study employs a whole-institution action research approach spanning policy, program, and course levels, combining analysis of a top-down policy intervention, qualitative discourse analysis of program evaluations across the institution, and development and piloting of a novel framework with teachers in problem- and project-based courses. The findings reveal that wicked problems are widely acknowledged in the program evaluations, however, often reframed into more familiar concepts, leading to limited and implicit rather than systematic consideration, and that related learning progression through programs is assumed rather than explicitly designed for. The developed framework demonstrates analytical and practical value by enabling differentiation between different dimensions and levels of "wickedness" in problem- and project-based courses, across dimensions such as problem complexity, students and stakeholder diversity and relations, and sustainability. Overall, the study highlights the need and provides tools for deliberate pedagogical design and coordinated strategies at course, study program, and university levels, for enhancing the relevance of engineering education with regard to wicked problems.

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1 Introduction

Climate change, water-energy-food security and sustainability, and other contemporary societal challenges and predicaments (e.g. as framed by the SDGs, UN 2015) are obviously not purely technical but involve interdependent social, environmental, economic, political, and other, dimensions. Engineering is therefore increasingly expected to contribute to systemic transformations rather than isolated technical solutions (UNESCO 2021). Traditional engineering education curricula, however, tend to emphasize well-structured, decontextualized problems with clear solution paths, reinforcing reductionist approaches to problem-solving (Jonassen et al., 2006; Lönngren et al., 2017).

Wicked problems, characterized by ambiguity, uncertainty, multiple stakeholders with conflicting interests and values, and the absence of definitive solutions, has gained traction as a concept for framing societal challenges (Rittel & Webber, 1973; Head, 2019). Its theoretical status, however, remains contested and the term 'wicked problem' is used interchangeably with terms such as 'intractable problem' and 'complex problem' with diverse meanings and epistemological assumptions, and commonly more as a rhetorical label than an analytical framing and tool for action (Lönngren & van Poeck 2021; Alford & Head 2017).

As pedagogical responses to these challenges, various forms of problem- and project-based, challenge-based, and challenge-driven, teaching and learning approaches have emerged (e.g. Wiek et al 2014; Kohn Rådberg et al 2018; Högfeldt et al 2019). These are experiential and action-oriented approaches engaging students, preferably from different disciplines, with complex real world societal challenges in collaboration with various societal stakeholders (van den Beemt et al., 2022). As relatively new and still-evolving concepts they, however, might not always fulfil their promises, for example by reinforcing technocentric, employability-focused, and market-driven agendas while overlooking political, ethical, and power-related sustainability conflicts (Larsson et al., 2025). Their definitional ambiguity and varied implementations are further raised as problematic (Gallagher & Savage, 2020) while, at the same time, their diversity and multivocality can be seen as enriching (Rosén et al., 2022).

Hence, while there is growing agreement on the need to develop engineering students' and engineers' capabilities to navigate complex societal challenges and their agency in sustainability transitions, there remains a lack of clarity regarding how key concepts are defined, operationalized, and translated into purposeful educational practices. With this background and with a specific Northern European technical university as context, tracing the concept of wicked problems from institutional policy, through program-level evaluation, to course-level implementation, this study explores the following research questions:

RQ1: How are wicked problems currently perceived and implemented in different educational programs?
RQ2: How can wicked problems be nuanced and concretized in problem- and project-based courses?

It should be noticed that in this paper 'problem- and project-based courses' is used as an overarching term also including courses labelled as project-based, problem-

based, challenge-based and challenge-driven, while being aware of their potential differences and diverse and ambiguous conceptualisations and implementations. The paper contributes to engineering education research as well as practice by: i) providing insights in an institution-wide evaluation of how wicked problems are perceived and implemented in educational programs; ii) synthesising theoretical and practical perspectives from the ongoing wicked problem discourse in the development of a qualitative framework for characterising wickedness in problem- and project-based courses; and iii) demonstrating and discussing the functionality of the framework in applications on existing courses.

2 Methodology

With the multiple intentions of doing research while developing teachers' understandings, pedagogical practices, and institutional strategies, this study resembles an action research process (Laudonia et al 2018). Ideally, action research can be described as a spiral of planning a change, action to implement the change, observing the process and consequences of the change, reflection, and continuing cycles of re-planning, action, reflection, etc. However, the process is generally more fluid, open and responsive, where the different stages overlap and initial plans might become successively obsolete in the light of generated learning (Kemmis et al 2014). Action research is commonly targeting one specific object; in education it can for example be one specific course or module. Unique with this study is, however, that it considers a whole-institution approach to enhancing the relevance of engineering education with regards to wicked problems, spanning the whole system from top-down policy interventions, through program evaluations, faculty development, to bottom-up perspectives on implementation details in separate courses. The study's cycles are described in the following where it can be noticed that the first cycle was more top-down and not initiated as an action research intervention, while the following cycles evolve towards participatory action research where teachers, educational researchers, and managers, collaborate.

2.1 Cycle 1, primarily concerning the first research question

The study can be seen to have its starting point in 2022 when the university introduced "skills to handle wicked problems for a sustainable societal development" as a new prioritised element in the internal educational policy. Recalling the definitional ambiguity highlighted in the introduction, it can be noticed that the term 'wicked problems' was used in the university's English version of this policy, while the term used in the native language version would be more directly translated to 'intractable problems'.

It had not been clarified to what extent and how wicked problems were considered in the educational programs before 2022, however, it is reasonable to assume that it was not that extensive. In 2024 the following questions concerning wicked problems were formulated and implemented in the formal template for program analyses that all programs do biannually: Are there courses in the program where students practice dealing with wicked problems? Is there progression for wicked problems through the program? Do you think the students reach a sufficient level in terms of their ability to deal with wicked problems within the program? The program analyses are self-evaluations conducted by the program directors in dialogue with teachers and other stakeholders. The analyses consider multiple quantitative as well as

qualitative aspects of the programs, are reported in written form, and forms the basis for program development, internal quality assurance, and national accreditation.

In total 90 programs on bachelor and master levels (basically all programs at this university) responded to these new wicked problems related questions in their program analyses. An educational researcher decided to go deeper in these responses through a qualitative, interpretive discourse analysis (Gee, 2014). First, responses were read as a coherent dataset, interpreting each program's responses as a way of 'talking about' wicked problems. Then, recurring linguistic and thematic patterns were identified and thematised into broader discursive patterns, focusing on: how wicked problems were framed in the programs; which pedagogical formats were linked to them; how progression was described; and how students' learning was justified.

2.2 Cycle 2, primarily concerning the second research question

Two observations in the first cycle (see the result section 3.1) motivated and guided the planning of a second cycle of this study: i) The binary labelling of wicked problems seems to invite either adoption or rejection, rather than productive analytical and practical engagement; ii) Wicked problems were in many program analyses associated with problem- and project-based learning in general and several programs highlighted specific courses. It was therefore decided to engage deeper with the wicked problem literature and with teachers with problem- and project-based courses. For this purpose, a qualitative framework was constructed and implemented in a questionnaire.

The construction of the framework was inspired by the way in which van den Beemt et al (2022) are characterising CBL (challenge-based learning). However, while van den Beemt et al (2022) have the ambition to capture basically all aspects of CBL, we are here focusing on the 'wickedness' of the problems/challenges that are addressed in the student projects. In contrast to more original ways of characterising problems as either 'wicked' or 'tame' (Rittel & Webber 1973) we are making use Alford & Head (2017) for a more nuanced and separate characterisation of, on the one hand the complexity of the problem itself, and on the other hand the complexity of stakeholder relations. In addition, we are detailing the diversity of the engaged students and stakeholders, characterising different sustainability dimensions, global issues, norms and values, on different levels in relation to Bloom's taxonomy (Krathwohl 2002), and also consider behavioural change and entrepreneurial aspects.

A first version of the framework was drafted by two educational researchers and was then iteratively modified and refined in dialogue with a handful of teachers with problem- and project-based courses who were engaged as co-researchers. The principles of the resulting framework are outlined in Table 1. When implemented in a questionnaire a section concerning informed consent was also included. As a pilot the questionnaire was sent out to 60 teachers in December 2025 to January 2026. From the 19 received responses four courses with rather different characteristics were chosen and used for demonstrating and discussing the framework in the result section 3.2.

In this paper we are limiting the study to investigating the courses separately from programs they might be part of. Such connections will be considered when the study continues in following cycles as outlined in the discussion and conclusion section.

Table 1: Framework for characterising different dimensions and levels of 'wickedness' in student projects in problem- and project-based courses.

Dimensions	**Levels of wickedness**
a) Students' disciplinary diversity	1. Students from one program/discipline. 2. …from different programs within the same area (e.g. different engineering programs). 3. …from different disciplines (e.g. engineering, humanities, and social sciences).
b) Problem character (levels adapted from Alford & Head 2017)	1. The problem/challenge as well as the major direction of the solution/project outcome are rather clear from the beginning of the project (e.g. designing a bridge for a certain location, or some component in an existing system or process). 2. The problem/challenge is relatively clear but might require some reframing in the beginning of the project. The solution/project outcome is however unclear and could develop in rather different directions and take different forms. ('Open-ended'). 3. The problem/challenge as well as the solution/project outcome are unclear, involving multiple dimensions and conflicting interests, and could develop in very different directions and take very different forms. ('Open-ended' as well as 'open-entry' and 'open-process").
c) Stakeholder engagement	1. Stakeholders are considered theoretically and/or hypothetically but not interacted with (e.g. considered through policies or other documents, or students' speculations or role plays). 2. One or a couple of key stakeholders are engaged and interacted with 'for real' (e.g. the problem/challenge provider is communicated with, visited or invited to some meetings) and yet some more might be considered theoretically and/or hypothetically. 3. Several stakeholders are engaged and interacted with 'for real' and yet some more might be considered theoretically and/or hypothetically.
d) Stakeholder diversity	Concerns from which societal sectors stakeholders are engaged - *private sector*, *public sector*, *civil society*, *policy organisations*, *research institutions*. In the results section the number of different sectors are presented in normalized form where 0.6 corresponds to engagement from one sector and 3 corresponds to engagement from all five sectors.
e) Stakeholder relations (levels adapted from Alford & Head 2017)	1. Tractable: considered stakeholders basically agree on the problem/challenge, possible solutions, and their different roles in relation to the project (e.g. designer-user/customer). 2. Diverse: considered stakeholders are broadly in consensus or at least indifferent about the nature of the problem/challenge and possible solutions, they are however more diverse in where they come from, what roles they have, what they might contribute with, and how they might benefit. 3. Conflicting: considered stakeholders have very different and conflicting views on and interests in the problem/challenge and possible solutions; power imbalances and underrepresented stakeholders (e.g. people in other places, future generations, ecosystems, other species) are thoroughly considered.
f-j) Ecological, social, economic sustainability, global issues, norms & values	1. Briefly (e.g. discuss but not analyse). 2. To some extent (e.g. analyse, with some influence on project outcomes). 3. Extensively (e.g. analyse, critically evaluate, relate to other sustainability aspects, and thoroughly implement in project outcomes). (Levels related to Bloom's taxonomy, Krathwohl 2002)
k) Behaviour change (levels related to Sterling 2011)	To what extent the students' projects potentially could change the behaviours of those who might use the solutions/project outcomes: 1. Briefly (e.g. related to changing practices and procedures). 2. To some extent (e.g. as in 2 and additionally related to changes in assumptions and norms). 3. Extensively (e.g. as in 3 and additionally related to changes in values, beliefs, and maybe even worldviews).
l) Entrepreneurial issues	To what extent students are expected to consider entrepreneurial issues in the projects: 1. Briefly (e.g. students develop business plans as part of the project outcomes). 2. To some extent (e.g. students develop plans for further development and implementation as part of the project outcomes). 3. Extensively (e.g. students develop start-ups or other ventures as part of the project outcomes).

3 Results

3.1 How wicked problems are perceived in programs

The discourse analysis in the first cycle of this study, of the program analyses replies to the wicked problem related questions, reveals several recurring patterns. A few programs frame wicked problems in line with the somewhat common understanding of wicked problems in the literature, for example linked to aspects such as uncertain knowledge base, inter- and transdisciplinarity, stakeholder diversity, and conflicting interest and values. For many programs, however, a key pattern is a noticeable distancing from the term itself. While many respondents acknowledge the phenomenon, they question or reframe the label. Statements such as “there are no problems in the program that are called wicked problems” or that the term is “new and trendy” illustrate this. Instead, respondents frequently translate the concept into more familiar categories, such as “open-ended problems” or simply any “engineering problem”. This suggests that the phenomenon is accepted insofar as it aligns with established educational vocabularies, rather than as a distinct conceptual category. The normalization of wicked problems as inherent to the discipline or professional field is another observed pattern. Several program analyses frame such problems as intrinsic to areas like sustainability and urban development, and others to engineering practice more broadly, as “by definition” involving wicked problems. This normalization legitimizes existing practices by implying that such problems are already addressed. However, it also risks limiting explicit curricular development, as the phenomenon is assumed to be omnipresent rather than intentionally considered. Yet another pattern is the association of wicked problems with specific pedagogical formats, such as problem- and project-based courses, studio settings, and degree projects. These contexts are characterized by open-ended questions, incomplete information, and often involving external stakeholders. Wicked problems are thus constructed less as a defined body of knowledge and more as something that emerges through particular teaching practices. Closely related is another pattern concerning progression. Learning through programs are typically described as a gradual movement from well-defined to open-ended problems, from structured guidance to independence, and from foundational knowledge to skills, positioning wicked problems in later stages of the programs. This reflects a traditional educational logic in which disciplinary knowledge is seen as a prerequisite for engaging with complexity. At the same time, many program analyses explicitly note that progression is unclear, implicit, or uneven. Statements such as “there is no clear progression” or “it exists, but not systematically” indicate self-critique. Even where wicked problems are considered, it often depends on individual courses or teachers rather than being integrated in the program design. Finally, programs tend to justify students’ learning through indirect indicators such as degree project quality, employability, and relevance to industry, rather than based on formalized assessment.

3.2 Wicked problems in problem- and project-based courses

As mentioned in the method section, the second cycle of the study enabled the identification of four courses of rather different character which are here used for demonstrating and discussing the framework. All four courses are on MSc level and have the students’ engagement in projects as the dominating learning activity, in course 1 in teams of 2 and in courses 2-4 in teams of 4-8 students. The questionnaire outcomes for these courses are presented in Figure 1. Recalling the

framework in Table 1, and that the levels 1-3 are intended to capture increasing 'wickedness', it can be argued that the problems addressed in course 1 are the least wicked, course 4 address the most wicked problems, whereas courses 2 and 3 are in between. The questionnaire allowed multiple (and omitted) answers to each dimension, and the dotted lines indicate where the teachers have utilised this opportunity. For courses 2 and 4 the teachers clarify that these optional replies reflect that the projects might differ between different student teams and between different course rounds. For course 1 the teacher describes that the optional replies reflects that these two dimensions are considered on higher levels (2-3) in discussions and role plays while still not being thoroughly considered in the project outcomes. Course 1 is a project-based course where the students are designing some parts of a technical system. Originally the course and the projects were only considering the technical aspects. However, the teacher has developed the course for example by extending the systems boundaries, considering a wider range of stakeholders, and environmental as well as social sustainability. In course 2 the student projects are centred around product development for an external stakeholder. Also here the teachers have gradually developed the course to consider other stakeholder categories and various sustainability dimensions. A key project process model that the students use in this course is the triple-layered business model canvas, which might be reflected in the high levels (2-3) of economic sustainability. Courses 3 and 4 are both transdisciplinary, where very diverse student groups address highly complex and open-ended problems in close collaboration with, in the case of course 3 one public sector stakeholder and in the case of course 4 a multitude of stakeholders from all sectors of society. Key project process models in these courses are design thinking in course 3 and backcasting in course 4.

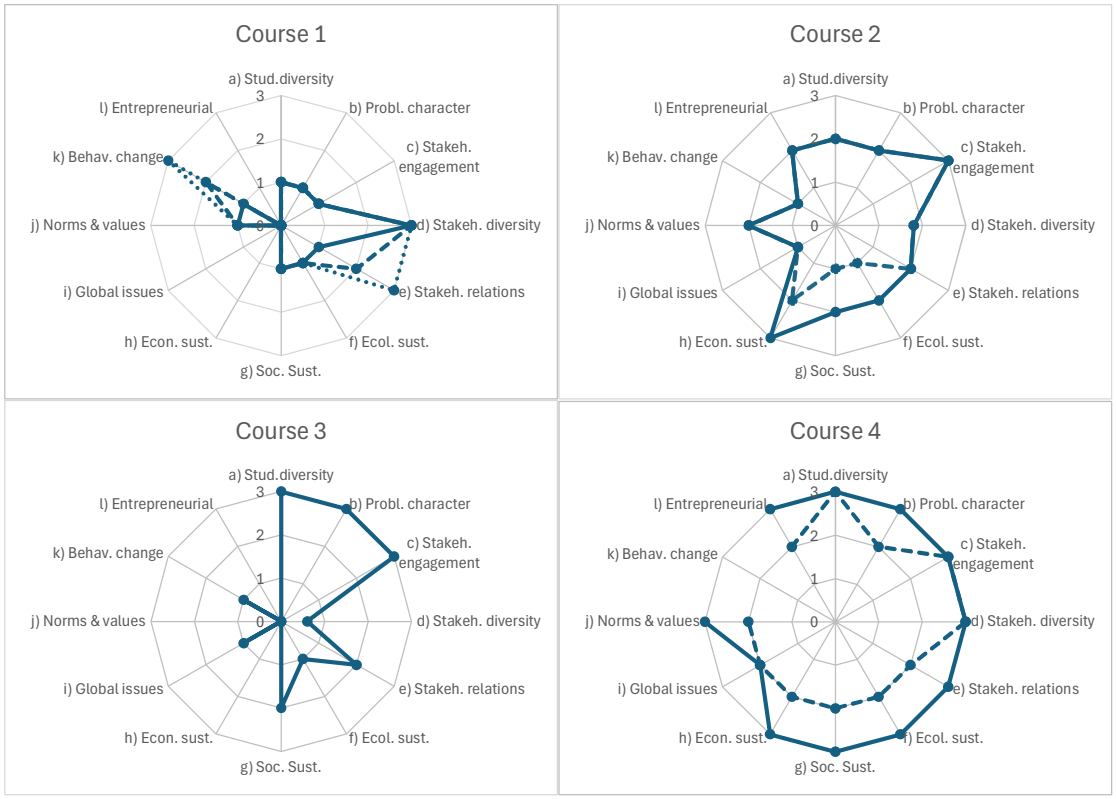


Figure 1: Survey outcomes for the four studied problem- and project-based courses.

4 Discussion and conclusions

The findings from the first cycle of this study, concerning the first research question on how wicked problems are perceived and implemented in educational programs, reveal a clear tension between recognition and operational clarity. While many programs acknowledge the relevance of wicked problems, they often reinterpret them through more familiar concepts such as open-ended problems or argue that any engineering problem is wicked, thereby diluting their distinct epistemic and pedagogical implications. This supports earlier critiques that the notion of wickedness risks becoming rhetorical rather than analytical when not explicitly defined and implemented. The findings further highlight that programs commonly argue that wicked problems are embedded implicitly within existing pedagogical formats. The association of wicked problems with later stages of programs reflects a traditional progression logic, where complexity is introduced after disciplinary foundations are established. However, the lack of explicit progression, as identified across programs, suggests that students' development of capabilities to navigate and take action in contexts of wicked problems may be uneven and dependent on individual courses or educators. This indicates the needs for and opportunities in more deliberate consideration of wicked problems in program designs.

As a response to the findings in the first cycle, and more specifically to the second research questions, the second cycle of this study has concerned the development and testing of a framework for characterising different dimensions and levels of 'wickedness' in student projects in problem- and project-based courses. The framework's analytical value is demonstrated in differentiating problem complexity, student and stakeholder diversity and relations, sustainability and other systems perspectives, between different courses. The framework is intended as a reflective and strategic tool for course and curriculum development, as well as for mutual learning and the exchange of experiences and ideas. Findings from the second cycle indicate that the framework's more granular and gradable categorisation of wickedness and its properties can help teachers move beyond binary distinctions, rhetorical labelling and disciplinary comfort zones, and enable them to describe their courses more precisely and make deliberate, visible choices about the degree of wickedness to incorporate. This can in turn support more nuanced collegial discussions about the contribution by different courses and how students could progressively develop capabilities for managing different aspects of wickedness through a programme.

The study will continue through subsequent cycles. First, the framework will be revised based on insights from the pilot study and through further refinement of its theoretical foundation (e.g. regarding correlations between different dimensions). An updated questionnaire will then be distributed to a larger group of teachers involved in problem- and project-based courses at the Northern European university where the present study was conducted and at a few partner universities. These teachers will subsequently be invited to focus-group workshops to discuss and refine the framework, explore interpretations of wickedness in their respective courses, and to jointly identify opportunities for course development and continued collaboration. In later stages, the study will examine how the framework and insights from these courses can be used to strengthen the integration of wickedness across educational programmes and to inform strategic development at programme, university, and cross-university levels.

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